

JKBOSE 2026

CLASS 11TH

**LIMITS AND
DERIVATIVES**

MATHS

LECTURE-5

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TOPICS TO BE COVERED

1

Introduction to Derivatives

2

Related Problems

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

$$\lim_{x \rightarrow 0} \frac{\ln x}{x} = 1.$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

QUESTION

Evaluate the following limits in Exercises 1 to 22. $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$

$$\lim_{x \rightarrow \pi/2} \frac{\tan 2x}{x - \frac{\pi}{2}}$$

(Sol) let $y = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan 2x}{x - \frac{\pi}{2}}$ $\left| \frac{0}{0} \text{ form} \right|$

$$\Rightarrow y = \lim_{x \rightarrow \frac{\pi}{2}^+} \frac{\tan 2x}{x - \frac{\pi}{2}}$$

$$\Rightarrow y = \lim_{h \rightarrow 0} \frac{\tan 2\left(\frac{\pi}{2} + h\right)}{\cancel{\frac{\pi}{2}} + h - \cancel{\frac{\pi}{2}}}$$

$$\Rightarrow y = \lim_{h \rightarrow 0} \frac{\tan(\pi + 2h)}{h}$$

$$\Rightarrow y = \lim_{h \rightarrow 0} \frac{\tan 2h}{2h} \times 2$$

$$\Rightarrow y = 2 \lim_{2h \rightarrow 0} \frac{\tan 2h}{2h} = 2 \times 1$$

$$\Rightarrow \boxed{y = 2} \text{ Hence the result.}$$

QUESTION

Find $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 1} f(x)$, where $f(x) = \begin{cases} 2x + 3, & x \leq 0 \\ 3(x + 1), & x > 0 \end{cases}$

(Sol) Here $\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} (3(h+1)) = 3.$

& $\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} (-2h+3) = 3.$

$\Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = 3.$

$\Rightarrow \lim_{x \rightarrow 0} f(x) = 3$

Also, $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} 3(x+1)$

$\Rightarrow \lim_{x \rightarrow 1} f(x) = 3(1+1) = 3 \times 2 = 6.$

$\Rightarrow \lim_{x \rightarrow 1} f(x) = 6.$

QUESTION

Find $\lim_{x \rightarrow 1} f(x)$ where $f(x) = \begin{cases} x^2 - 1, & x \leq 0 \\ -x^2 - 1, & x > 0 \end{cases}$

$$\text{H/W: } \lim_{x \rightarrow 1} (-x^2 - 1) = -2$$

QUESTION

Evaluate $\lim_{x \rightarrow 0} f(x)$, where $f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

(Sol) Given function is: $f(x) = \begin{cases} \frac{|x|}{x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$

$$\Rightarrow f(x) = \begin{cases} \frac{x}{x}; & x > 0 \\ 0; & x = 0 \\ -\frac{x}{x}; & x < 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} 1; & x > 0 \\ 0; & x = 0 \\ -1; & x < 0 \end{cases}$$

$$\text{Here } \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} (1) = 1.$$

$$\neq \lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} (-1) = -1.$$

$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

This shows that $\lim_{x \rightarrow 0} f(x)$ does not exist.

Hence the result.

QUESTION



Find $\lim_{x \rightarrow 0} f(x)$, where $f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

H/W.

QUESTION

Find $\lim_{x \rightarrow 0} f(x)$, where $f(x) = |x| - 5$

H/w.

$$f(x) = \begin{cases} x - 5, & x \geq 0 \\ -x - 5, & x < 0 \end{cases}$$

QUESTION

$$f(x) = \begin{cases} |x| + 1, & x < 0 \\ 0, & x = 0 \\ |x| - 1, & x > 0 \end{cases}$$

For what value (s) of a does $\lim_{x \rightarrow a} f(x)$ exists?

HP/W.
 $\lim_{x \rightarrow 0^+} f(x), \lim_{x \rightarrow 0^-} f(x)$

QUESTION

If the function $f(x)$ satisfies $\lim_{x \rightarrow 1} \frac{f(x) - 2}{x^2 - 1} = \pi$, evaluate $\lim_{x \rightarrow 1} f(x)$.

H/W

QUESTION

If $f(x) = \begin{cases} mx^2 + n, & x < 0 \\ nx + m, & 0 \leq x \leq 1 \\ nx^3 + m, & x > 1 \end{cases}$. For what integers m and n

does both $\lim_{x \rightarrow 0} f(x)$ & $\lim_{x \rightarrow 1} f(x)$ exist?

(Sol) Since $\lim_{x \rightarrow 0} f(x)$ exists, then $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$

$$\Rightarrow \lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} f(-h)$$

$$\Rightarrow \lim_{h \rightarrow 0} (nh + m) = \lim_{h \rightarrow 0} (m(-h)^2 + n)$$

$$\Rightarrow nx_0 + m = mx_0 + n$$

$$\Rightarrow \boxed{m = n}$$

This shows that $\lim_{x \rightarrow 0} f(x)$ exists whenever $m = n$.

Also, we have been given that $\lim_{x \rightarrow 1} f(x)$ exists, then

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} f(1-h)$$

$$\Rightarrow \lim_{h \rightarrow 0} [n(1+h)^3 + m] = \lim_{h \rightarrow 0} [n(1-h) + m]$$

$$\Rightarrow \boxed{n + m = n + m}$$

This shows that $\lim_{x \rightarrow 1} f(x)$ exists for all the values of m and n .

Hence the result.

Derivatives

★ **Definition:** Suppose that f be any real value function and a be any point in the domain of f . Then f is said to be differentiable at a point $x=a$

if $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists finitely. It's denoted

by $f'(a)$. So $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$.

$$* \lim_{x \rightarrow a^+} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad (\text{Right hand derivative})$$

$$x - a = \cancel{a} + h - \cancel{a} = h \rightarrow Rf'(a)$$

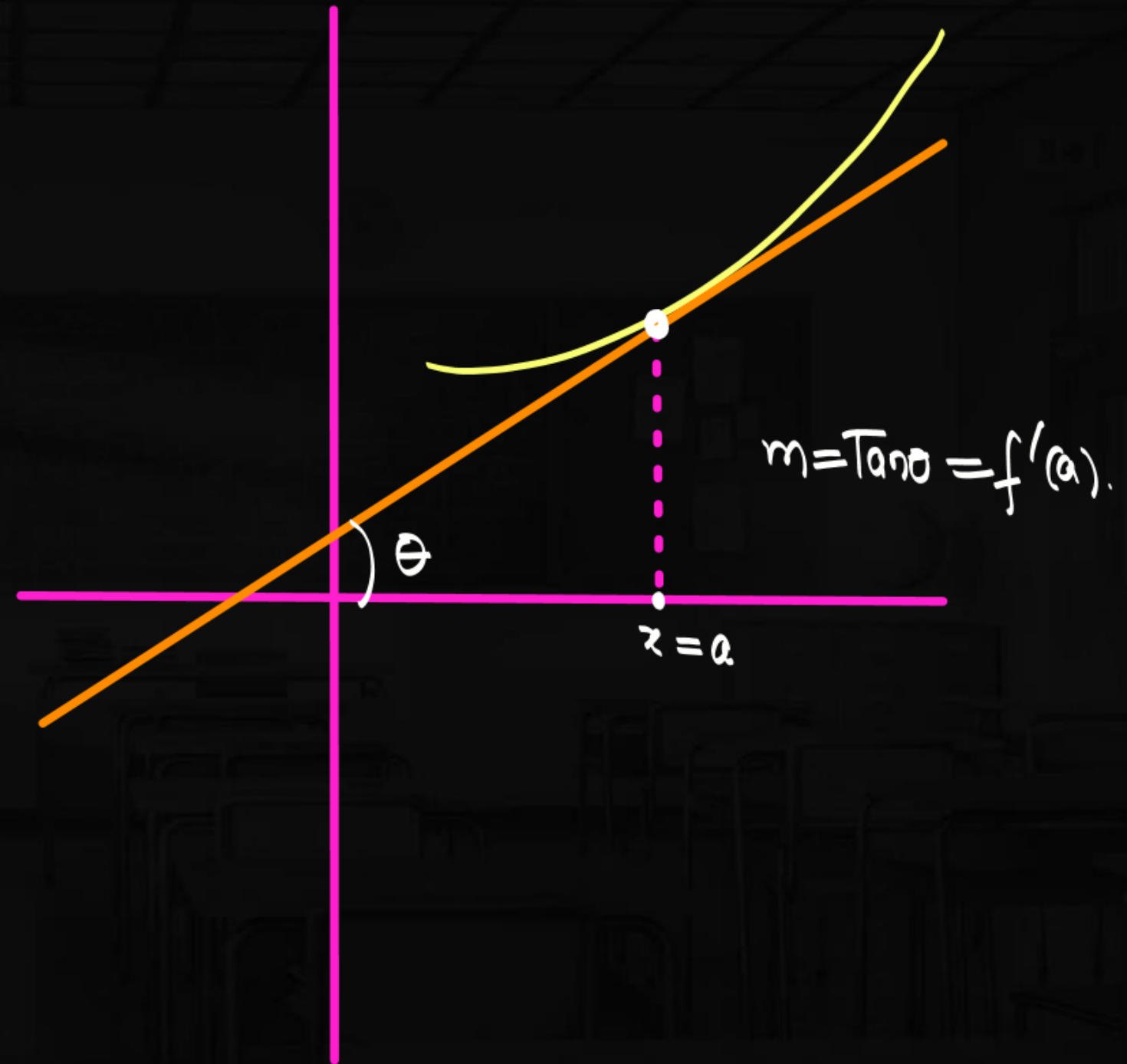
$$* \lim_{x \rightarrow a^-} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h} \quad (\text{left hand Derivative})$$

★ A function f is differentiable/derivable at a point $x=a$ if $Rf'(a) = Lf'(a)$. And their common value is denoted by $f'(a)$.

$$\text{So, } f'(a) = Rf'(a)$$

$$\Rightarrow f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad (\text{Derivative by first principle / ab-initio Method})$$

* Geometrically $f'(a)$ represents the slope of tangent to the curve $y = f(x)$ at $x = a$.



★ **Derivative of $x^n, n \in \mathbb{Z}^+$** : Suppose that $f(x) = x^n$ be the given function. Then by first principle (**ab-initio Method**), we get;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \quad \left| \begin{array}{l} \text{0} \\ \text{0 form} \end{array} \right.$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{n_0 x^n + n_1 x^{n-1} h + n_2 x^{n-2} h^2 + \dots + n_n h^n - x^n}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\cancel{x^n} + n x^{n-1} h + n_2 x^{n-2} h^2 + \dots + h^n - \cancel{x^n}}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{h \left(n x^{n-1} + n_2 x^{n-2} h + n_3 x^{n-3} h^2 + \dots + h^{n-1} \right)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left(n x^{n-1} + n_2 x^{n-2} h + n_3 x^{n-3} h^2 + \dots + h^{n-1} \right)$$

$$\Rightarrow f'(x) = n x^{n-1} + 0 + 0 + \dots + 0.$$

$$\Rightarrow \boxed{f'(x) = n x^{n-1}} \quad (\text{Power rule})$$

$$\rightarrow \frac{d}{dx} x^n = n x^{n-1}$$

EXAMPLE

$$(x^n)' \rightarrow nx^{n-1} \\ (x^1)' = 1 \cdot x^{1-1} = 1 \cdot x^0 = 1.$$

$$\frac{d}{dx}(a f(x)) = a \frac{d}{dx}(f(x)).$$

Find the derivative at $x = 2$ of the function $f(x) = 3x$.

(Sol) Given function is;

$$f(x) = 3x.$$

$$\Rightarrow f'(x) = 3 \times 1$$

$$\Rightarrow f'(x) = 3.$$

$$\Rightarrow \boxed{f'(2) = 3}.$$

By first principle,

$$f'(2) = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h}$$

$$\Rightarrow f'(2) = \lim_{h \rightarrow 0} \frac{3(2+h) - 3 \times 2}{h}$$

$$\Rightarrow f'(2) = \lim_{h \rightarrow 0} \left(\frac{\cancel{6} + 3h - \cancel{6}}{h} \right) = \lim_{h \rightarrow 0} \frac{3\cancel{h}}{\cancel{h}}$$

$$\Rightarrow f'(2) = \lim_{h \rightarrow 0} (3) = 3$$

$$\Rightarrow \boxed{f'(2) = 3}.$$

EXAMPLE

Find the derivative of the function $f(x) = 2x^2 + 3x - 5$ at $x = -1$.

Also prove that $f'(0) + 3f'(-1) = 0$

Proof: Given function is:

$$f(x) = 2x^2 + 3x - 5$$

$$\Rightarrow f'(x) = 2 \times 2x + 3 \times 1 - 0$$

$$\Rightarrow f'(x) = 4x + 3$$

$$\text{So, } f'(-1) = 4(-1) + 3 = -1$$

$$\Rightarrow f'(-1) = -1$$

$$\text{Also } f'(0) = 4 \times 0 + 3 = 3$$

$$\Rightarrow f'(0) = 3$$

Now, Consider the following;

$$f'(0) + 3f'(-1) = 3 + 3 \times -1 = \cancel{3} - \cancel{3}$$

$$\Rightarrow f'(0) + 3f'(-1) = 0$$

Hence proved.

Thank
You